PRACTICE QUESTIONS PROBABILITY TREES

EXERCISE – 2.9

PROBABILITY TREES

1. Tom travels to work either by bus or by train or by bicycle. The probabilities that he travels by bus is 0.3, by trains is 0.5 and by bicycle is 0.2. the probability that he arrives on time is 0.6 if he travels by bus, 0.9 if he travels by train and 0.8 if he travels by bicycle.
(i) Draw a tree diagram to show this information.
(ii) Find the probability that Tom does not arrive at work on time.
(iii) Find the conditional probability that Tom travelled by bus, given that he does not arrive at work on time.
$(Ans: 0.21, 0.571)$

2. The people living in 3 houses are classified as children (C), parents (P) or grandparents (G). The numbers living in each house are shown in the table below.

House number 1	House number 2	House number 3
$4C,\ 1P,\ 2G$	$2C,\ 2P,\ 3G$	$1C,\ 1G$

(i) All the people in all 3 houses meet for a party. One person at the party is chosen at random. Calculate the probability of choosing a grandparent.
(ii) A house is chosen at random. Then a person in that house is chosen at random. Using a tree diagram, or otherwise, calculate the probability that the person chosen is a grandparent.
(iii) Given that the person chosen by the method in part **(ii)** is a grandparent, calculate the probability that there is also a parent living in the house.
$(Ans: 3/8, 17/42, 10/17)$

3. In a certain country 54% of the population is male. It is known that 5% of the males are colour-blind and 2% of the females are colour-blind. A person is chosen at random and found to be colour-blind. By drawing a tree diagram, or otherwise, find the probability that this person is male.
$(Ans: 0.7458)$

4. When Don plays tennis, 65% of his first serves go into the correct area of the court. If the first serve goes into the correct area, his chance of winning the point is 90%. If his first serve does not go into the correct area, Don is allowed a second serve, and of these, 80% go into the correct area. If the second serve goes into the correct area, his chance of winning the point is 60%. If neither serve goes into the correct area, Don loses the point.
(i) Draw a tree diagram to represent this information.
(ii) Using your tree diagram, find the probability that Don loses the point.
(iii) Find the conditional probability that Don's first serve went into the correct area, given that he loses the point.
$(Ans: 0.247, 0.263)$

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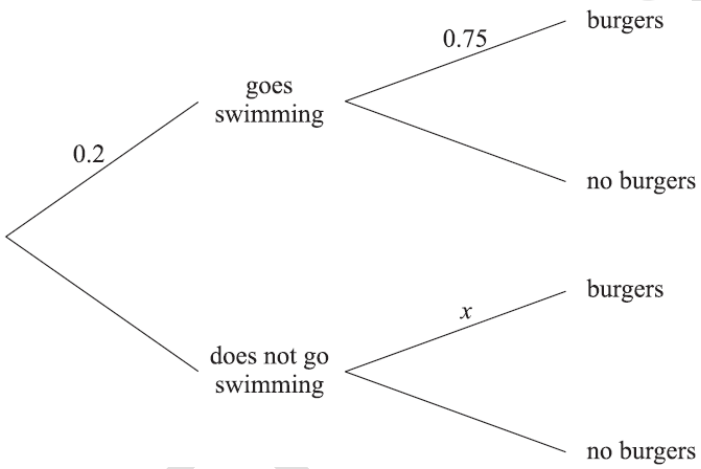
5. When Andrea needs a taxi, she rings one of three taxi companies, *A*, *B* or *C*. 50% of her calls are to taxi company *A*, 30% to *B* and 20% to *C*. A taxi from company *A* arrives late 4% of the time, a taxi from company *B* arrives late 6%of the time and a taxi from company *C* arrives late 17% of the time.

(i) Find the probability that, when Andrea rings for a taxi, it arrives late.

(ii) Given that Andrea's taxi arrives late, find the conditional probability that she rang company *B*.

$(Ans: 0.072, 0.250)$

6. The probability that Henk goes swimming on any day is 0.2. On a day when he goes swimming, the probability that Henk has burgers for supper is 0.75. On a day when he does not go swimming the probability that he has burgers for supper is *x*. This information is shown on the following tree diagram.



The probability that Henk has burgers for supper on any day is 0.5.

(i) Find *x*.

(ii) Given that Henk has burgers for supper, find the probability that he went swimming that day.

$(Ans: 0.4375, 0.3)$

7. There are three sets of traffic lights on Karinne's journey to work. The independent probabilities that Karinne has to stop at the first, second and third set of lights are 0.4, 0.8 and 0.3 respectively.

(i) Draw a tree diagram to show this information.

(ii) Find the probability that Karinne has to stop at each of the first two sets of lights but does not have to stop at the third set.

(iii) Find the probability that Karinne has to stop at exactly two of the three sets of lights.

(iv) Find the probability that Karinne has to stop at the first set of lights, given that she has to stop at exactly two sets of lights.

$(Ans: 0.224, 0.392, 0.633)$

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8. Fabio drinks coffee each morning. He chooses Americano, Cappucino or Latte with probabilities 0.5, 0.3 and 0.2 respectively. If he chooses Americano he either drinks it immediately with probability 0.8, or leaves it to drink later. If he chooses Cappucino he either drinks it immediately with probability 0.6, or leaves it to drink later. If he chooses Latte he either drinks it immediately with probability 0.1,or leaves it to drink later.
(a) Find the probability that Fabio chooses Americano and leaves it to drink later.
(b) Fabio drinks his coffee immediately. Find the probability that he chose Latte.
$(Ans: 0.1, 0.0333)$

9. On Saturday afternoons Ali goes shopping with probability 0.25, or goes to the cinema with probability 0.35 or stays at home. If he goes shopping the probability that he spends more than \$50 is 0.7. If he goes to the cinema the probability that he spends more than \$50 is 0.8. If he stays at home he spends \$10 on a pizza.
(a) Find the probability that Ali will go to the cinema and spend less than \$50.
(b) Given that he spends less than \$50, find the probability that he went to the cinema.
$(Ans: 0.07, 0.128)$

10. At a zoo, rides are offered on elephants, camels and jungle tractors. Ravi has money for only one ride. To decide which ride to choose, he tosses a fair coin twice. If he gets 2 heads he will go on the elephant ride, if he gets 2 tails he will go on the camel ride and if he gets 1 of each he will go on the
jungle tractor ride.
(i) Find the probabilities that he goes on each of the three rides.
The probabilities that Ravi is frightened on each of the rides are as follows:

$$\text{elephant ride } \frac{6}{10}, \quad \text{camel ride } \frac{7}{10}, \quad \text{jungle tractor ride } \frac{8}{10}.$$

(ii) Draw a fully labelled tree diagram showing the rides that Ravi could take and whether or not he is frightened.
Ravi goes on a ride.
(iii) Find the probability that he is frightened.
(iv) Given that Ravi is **not** frightened, find the probability that he went on the camel ride.
$(Ans: \frac{1}{4}, \frac{1}{4}, \frac{1}{2}, 0.725, 0.273)$

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PRACTICE QUESTIONS BAYE'S THEOREM

EXERCISE – 2.10

BAYE'S THEOREM

11. A rare disease exists with which only 1 in 500 is affected. A test for the disease exists, but of course it is not infallible. A correct positive result (patient actually has the disease) occurs 95% of the time, while a false positive result (patient does not have the disease) occurs 1% of the time. If a randomly selected individual is tested and the result is positive, what is the probability that the individual has the disease?
((*Ans* :0.1599)

12. A certain form of cancer is known to be found in women over 60 with probability 0.07. A blood test exists for the detection of the disease, but the test is not infallible. In fact, it is known that 10% of the time the test gives a false negative (i.e., the test incorrectly gives a negative result) and 5% of the time the test gives a false positive (i.e., incorrectly gives a positive result). If a woman over 60 is known to have taken the test and received a favorable (i.e., negative) result, what is the probability that she has the disease?
(*Ans* : 0.00786) Walpole 2.118

13. A truth serum has the property that 90% of the guilty suspects are properly judged while, of course,10% of the guilty suspects are improperly found innocent. On the other hand, innocent suspects are misjudged 1% of the time. If the suspect was selected from a group of suspects of which only 5% have ever committed a crime, and the serum indicates that he is guilty, what is the probability that he is innocent?
(*Ans* : 0.1743) Walpole 2.103

14. A paint-store chain produces and sells latex and semigloss paint. Based on long-range sales, the probability that a customer will purchase latex paint is 0.75. Of those that purchase latex paint, 60% also purchase rollers. But only 30% of semigloss paint buyers purchase rollers. A randomly selected buyer purchases a roller and a can of paint. What is the probability that the paint is latex?
(*Ans* : 0.857) Walpole 2.101

15. A firm is accustomed to training operators who do certain tasks on a production line. Those operators who attend the training course are known to be able to meet their production quotas 90% of the time. New operators who do not take the training course only meet their quotas 65% of the time. Fifty percent of new operators attend the course. Given that a new operator meets her production quota, what is the probability that she attended the program?
(*Ans* : 0.581) Walpole 2.124

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16. The members of a consulting firm rent cars from three rental agencies: 60 percent from agency 1, 30 percent from agency 2, and 10 percent from agency 3. If 9 percent of the cars from agency 1 need an oil change, 20 percent of the cars from agency 2 need an oil change, and 6 percent of the cars from agency 3 need an oil change.
(a) what is the probability that a rental car delivered to the firm will need an oil change?
(b) If a rental car delivered to the consulting firm needs an oil change, what is the probability that it came from rental agency 2?
$(Ans: 0.12,\ 0.5)$

17. A certain federal agency employs three consulting firms (*A*, *B*, and *C*) with probabilities 0.40, 0.35, and 0.25, respectively. From past experience it is known that the probability of cost overruns for the firms are 0.05, 0.03, and 0.15, respectively. Suppose a cost overrun is experienced by the agency.
(a) What is the probability that the consulting firm involved is company *C*?
(b) What is the probability that it is company *A*?
$(Ans: 0.5515,\ 0.2941)$

18. A producer of a certain type of electronic component ships to suppliers in lots of twenty. Suppose that 60% of all such lots contain no defective components, 30% contain one defective component, and 10% contain two defective components. A lot is picked, two components from the lot are randomly selected and tested, and neither is defective.
(a) What is the probability that zero defective components exist in the lot?
(b) What is the probability that one defective exists in the lot?
(c) What is the probability that two defectives exist in the lot?
$(Ans: 0.6312,\ 0.2841, 0.0847)$

19. Suppose that the four inspectors at a film factory are supposed to stamp the expiration date on each package of film at the end of the assembly line. John, who stamps 20% of the packages, fails to stamp the expiration date once in every 200 packages; Tom, who stamps 60% of the packages, fails to stamp the expiration date once in every 100 packages; Jeff, who stamps 15% of the packages, fails to stamp the expiration date once in every 90 packages; and Pat, who stamps 5% of the packages, fails to stamp the expiration date once in every 200 packages. If a customer complains that her package of film does not show the expiration date, what is the probability that it was inspected by John?
$(Ans: 0.1124)$

20. A regional telephone company operates three identical relay stations at different locations. During a one-year period, the number of malfunctions reported by each station and the causes are shown below.

Station	*A*	*B*	*C*
Problems with electricity supplied	2	1	1
Computer malfunction	4	3	2
Malfunctioning electrical equipment	5	4	2
Caused by other human errors	7	7	5

Suppose that a malfunction was reported and it was found to be caused by other human errors. What is the probability that it came from station *C*?
$(Ans: 0.2632)$

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SOLUTIONS OF PROBLEMS OF BAYE'S THEOREM

1. Consider events:
D: a person has the rare disease, $P(D) = 1/500$,
P: the test shows a positive result, $P(P \mid D) = 0.95$ and $P(P \mid D') = 0.01$.
$$P(D \mid P) = \frac{P(P \mid D)P(D)}{P(P \mid D)P(D)+P(P \mid D')P(D')} = \frac{(0.95)(1/500)}{(0.95)(1/500)+(0.01)(1-1/500)} = 0.1599.$$

2. Consider the events:
C: a woman over 60 has the cancer,
P: the test gives a positive result.
So, $P(C) = 0.07$, $P(P' \mid C) = 0.1$ and $P(P \mid C') = 0.05$.
$$P(C \mid P') = \frac{P(P' \mid C)P(C)}{P(P' \mid C)P(C)+P(P' \mid C')P(C')} = \frac{(0.1)(0.07)}{(0.1)(0.07)+(1-0.05)(1-0.07)} = \frac{0.007}{0.8905} = 0.00786.$$

3. Consider the events:
G: guilty of committing a crime,
I: innocent of the crime,
i: judged innocent of the crime,
g: judged guilty of the crime.
$$P(I \mid g) = \frac{P(g \mid I)P(I)}{P(g \mid G)P(G)+P(g \mid I)P(I)} = \frac{(0.01)(0.95)}{(0.05)(0.90)+(0.01)(0.95)} = 0.1743.$$

4. Consider the events:
A: a customer purchases latex paint,
A': a customer purchases semigloss paint,
B: a customer purchases rollers.
$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B \mid A)P(A)+P(B \mid A')P(A')} = \frac{(0.60)(0.75)}{(0.60)(0.75)+(0.25)(0.30)} = 0.857.$$

5. Consider the events:
T: an operator is trained, $P(T) = 0.5$,
M an operator meets quota, $P(M \mid T) = 0.9$ and $P(M \mid T') = 0.65$.
$$P(T \mid M) = \frac{P(M \mid T)P(T)}{P(M \mid T)P(T)+P(M \mid T')P(T')} = \frac{(0.9)(0.5)}{(0.9)(0.5)+(0.65)(0.5)} = 0.581.$$

6. **Solution**
If A is the event that the car needs an oil change, and B_1, B_2, and B_3 are the events that the car comes from rental agencies 1, 2, or 3, we have $P(B_1) = 0.60$, $P(B_2) = 0.30$, $P(B_3) = 0.10$, $P(A|B_1) = 0.09$, $P(A|B_2) = 0.20$, and $P(A|B_3) = 0.06$. Substituting these values into the Law of Total Probability formula, we get

$$P(A) = (0.60)(0.09) + (0.30)(0.20) + (0.10)(0.06)$$
$$= 0.12$$

(b) Substituting the probabilities on the previous page into the Baye's formula we get

$$P(B_2|A) = \frac{(0.30)(0.20)}{(0.60)(0.09)+(0.30)(0.20)+(0.10)(0.06)}$$
$$= \frac{0.060}{0.120}$$
$$= 0.5$$

7.

Consider the events:
O: overrun,
A: consulting firm A,
B: consulting firm B,
C: consulting firm C.

(a) $P(C \mid O) = \frac{P(O \mid C)P(C)}{P(O \mid A)P(A)+P(O \mid B)P(B)+P(O \mid C)P(C)} = \frac{(0.15)(0.25)}{(0.05)(0.40)+(0.03)(0.35)+(0.15)(0.25)} = \frac{0.0375}{0.0680} = 0.5515.$

(b) $P(A \mid O) = \frac{(0.05)(0.40)}{0.0680} = 0.2941.$

8.

Consider the events:
A: two nondefective components are selected,
N: a lot does not contain defective components, $P(N) = 0.6$, $P(A \mid N) = 1$,
O: a lot contains one defective component, $P(O) = 0.3$, $P(A \mid O) = \frac{\binom{19}{2}}{\binom{20}{2}} = \frac{9}{10}$,
T: a lot contains two defective components,$P(T) = 0.1$, $P(A \mid T) = \frac{\binom{18}{2}}{\binom{20}{2}} = \frac{153}{190}$.

(a) $P(N \mid A) = \frac{P(A \mid N)P(N)}{P(A \mid N)P(N)+P(A \mid O)P(O)+P(A \mid T)P(T)} = \frac{(1)(0.6)}{(1)(0.6)+(9/10)(0.3)+(153/190)(0.1)}$
$= \frac{0.6}{0.9505} = 0.6312;$

(b) $P(O \mid A) = \frac{(9/10)(0.3)}{0.9505} = 0.2841;$

(c) $P(T \mid A) = 1 - 0.6312 - 0.2841 = 0.0847.$

9.

Consider the events:
A: no expiration date,
B_1: John is the inspector, $P(B_1) = 0.20$ and $P(A \mid B_1) = 0.005$,
B_2: Tom is the inspector, $P(B_2) = 0.60$ and $P(A \mid B_2) = 0.010$,
B_3: Jeff is the inspector, $P(B_3) = 0.15$ and $P(A \mid B_3) = 0.011$,
B_4: Pat is the inspector, $P(B_4) = 0.05$ and $P(A \mid B_4) = 0.005$,
$P(B_1 \mid A) = \frac{(0.005)(0.20)}{(0.005)(0.20)+(0.010)(0.60)+(0.011)(0.15)+(0.005)(0.05)} = 0.1124.$

10.

Consider the events
E: a malfunction by other human errors,
A: station A, B: station B, and C: station C.
$P(C \mid E) = \frac{P(E \mid C)P(C)}{P(E \mid A)P(A)+P(E \mid B)P(B)+P(E \mid C)P(C)} = \frac{(5/10)(10/43)}{(7/18)(18/43)+(7/15)(15/43)+(5/10)(10/43)} = \frac{0.1163}{0.4419} = 0.2632.$